Chapter ML:III (continued)

III. Linear Models

- Logistic Regression
- Overfitting
- Regularization

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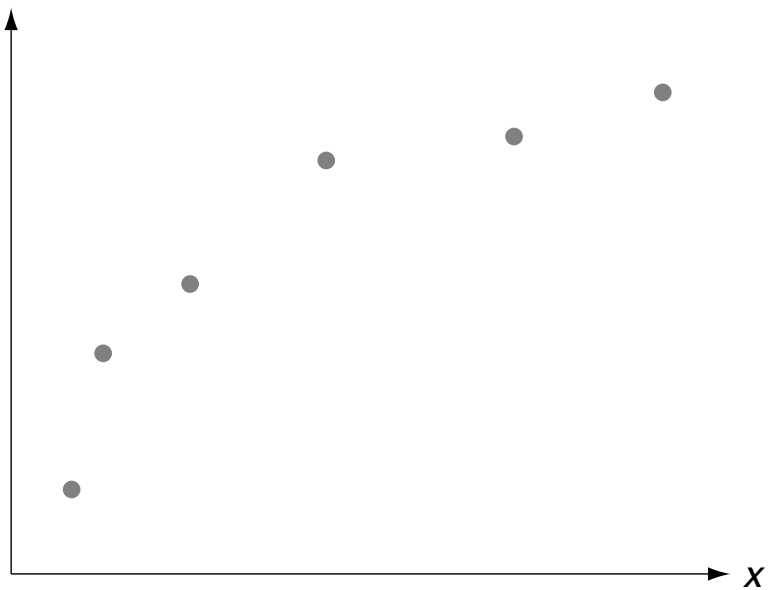
Overfitting

Example: Linear Regression



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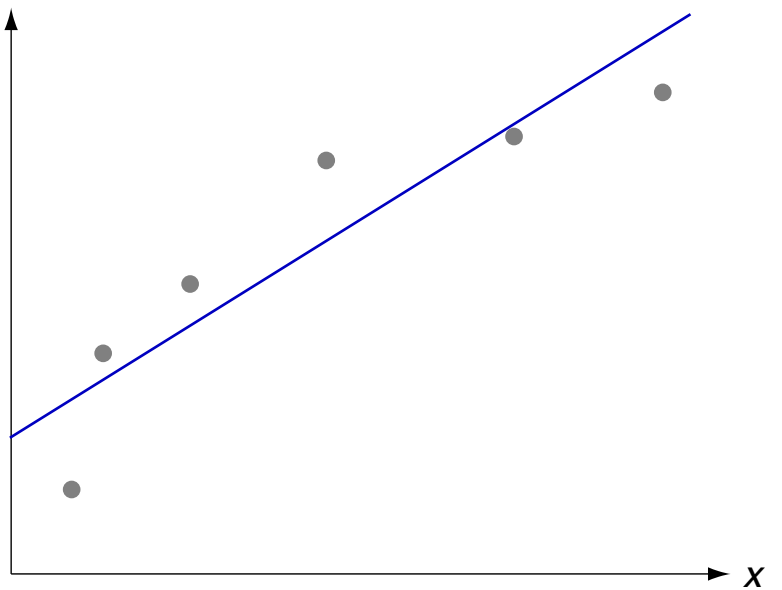
Overfitting

Example: Linear Regression



$$y(x) = w_0 + w_1 \cdot x$$

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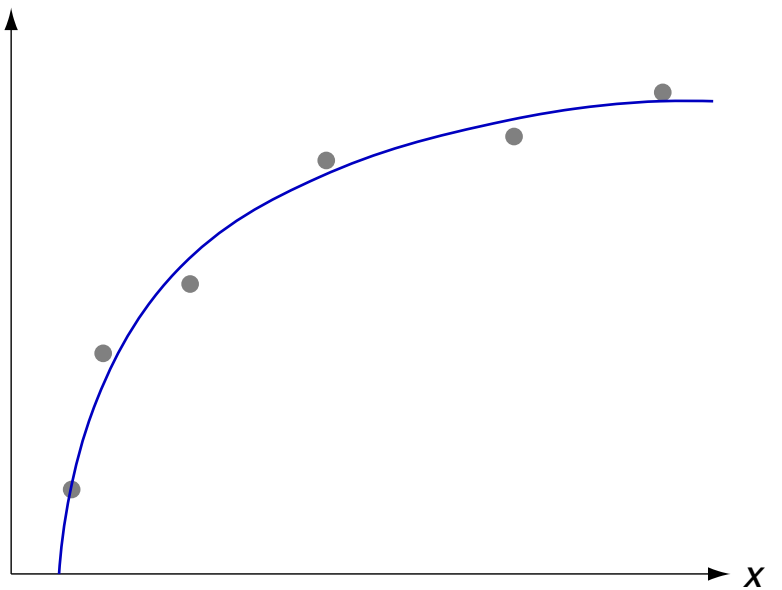
Overfitting

Example: Linear Regression



$$y(x) = w_0 + w_1 \cdot x + w_2 \cdot x^2$$

(basis expansion)

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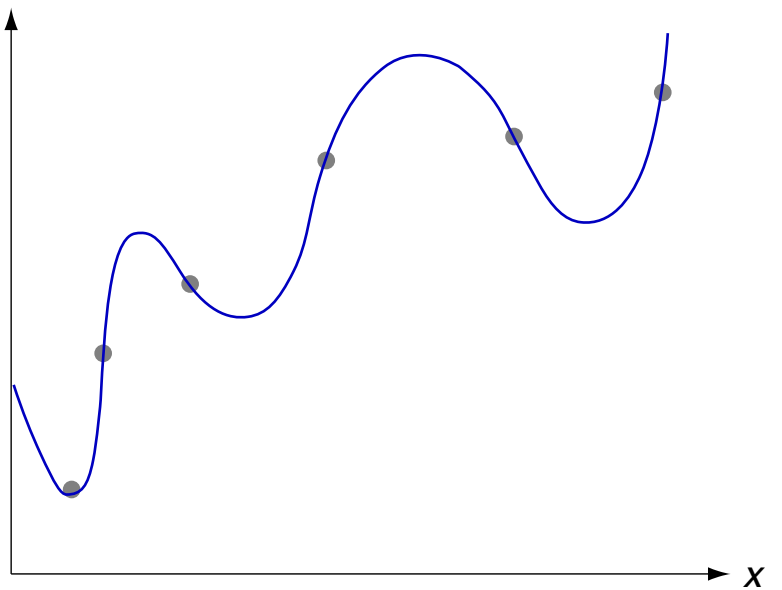
Overfitting

Example: Linear Regression



$$y(x) = w_0 + \sum_{i=1}^{6} w_i \cdot x^i$$

(basis expansion)

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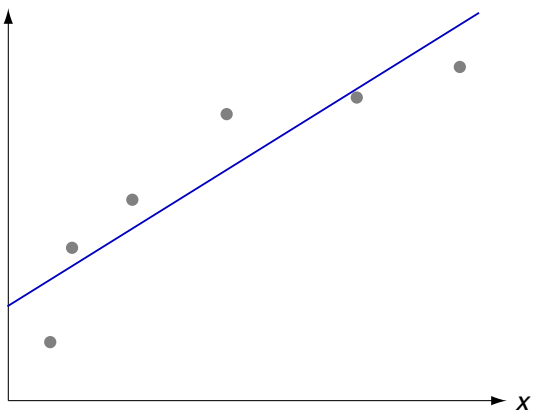
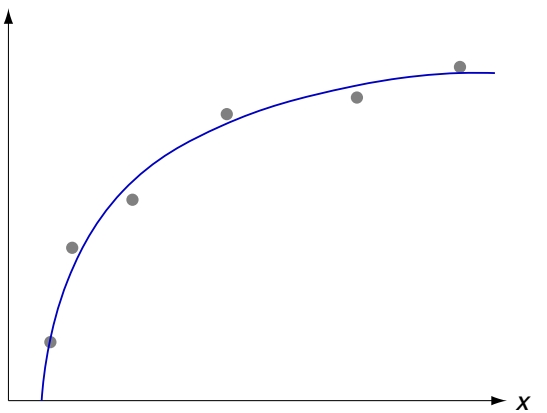
Overfitting

Example: Linear Regression

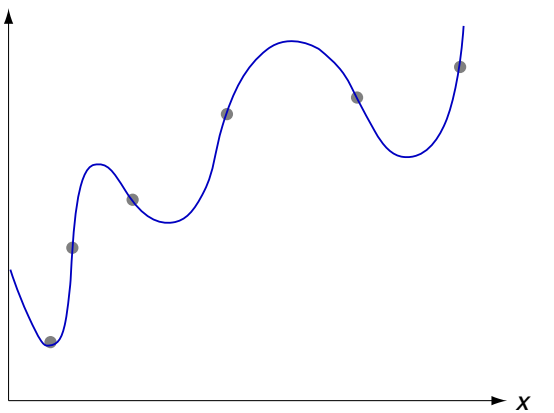
Given three polynomial model functions $y(x) = \mathbf{w}^T\mathbf{x}$ of degrees 1, 2, and 6, and a training set of D_{tr}, select the function that best fits the data:



RSS$(\mathbf{w}) \gg 0$



RSS$(\mathbf{w}) > 0$



RSS$(\mathbf{w}) = 0$

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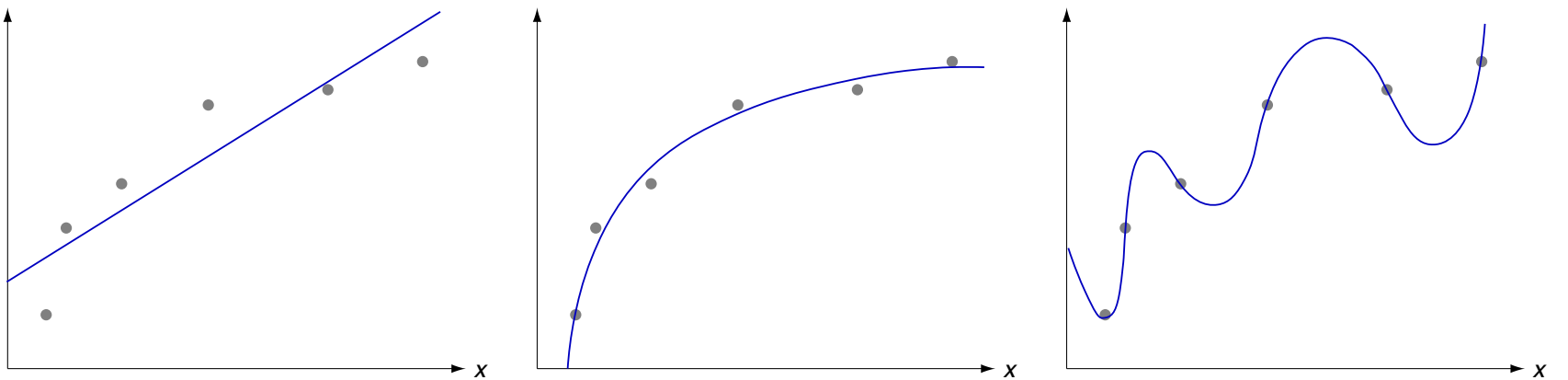

RSS$(\mathbf{w}) \gg 0$ RSS$(\mathbf{w}) > 0$ RSS$(\mathbf{w}) = 0$

Questions:

- Which model function (hypothesis) is the best choice?
- How to choose among the many hypothesis spaces available?

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Overfitting

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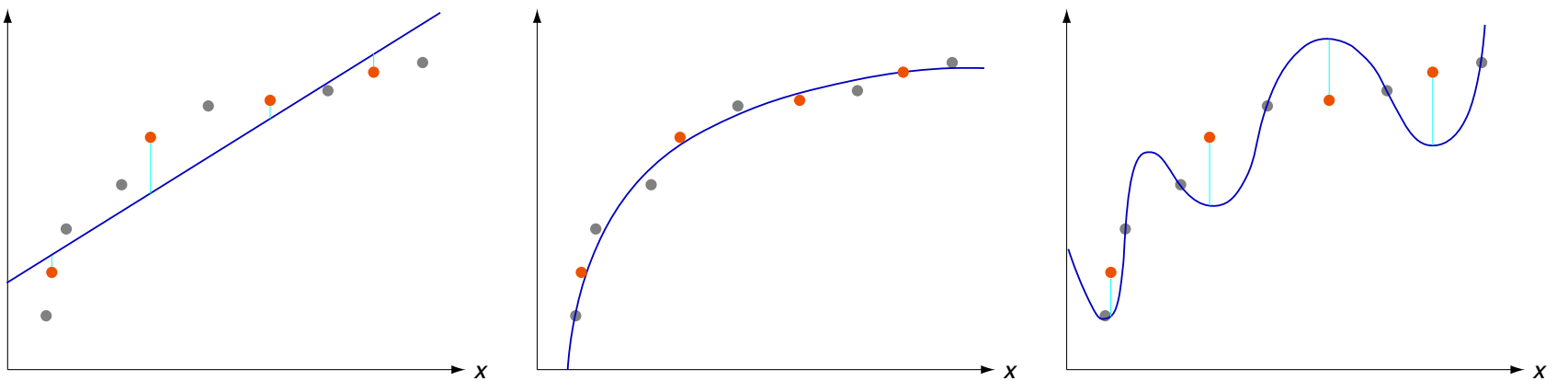

$\text{RSS}(\mathbf{w}) \gg 0$ $\qquad$ $\text{RSS}(\mathbf{w}) > 0$ $\qquad$ $\text{RSS}(\mathbf{w}) \gg 0$

Let D_{test} be a set of test examples.

If $D = D_{tr} \cup D_{test}$ is representative of the real-world population in X, the quadratic model function $y(x) = w_0 + w_1 \cdot x + w_2 \cdot x^2$ is the closest match.

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Overfitting

Definition

Model “fitting” is the process of optimizing the parameters of a model function y from a selected hypothesis space to obtain the “best fit” to a sample of data D.

Overfitting literally means “Fitting the data more than is warranted.” [Abu Mostafa, 2012]

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Causes:

- Too complex model function y (e.g., too many features, unrelated features).
- Too small sample of data D.
- Too much noise in the data (e.g., random fluctuations, errors).

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Effects:

- Low error on training data, but high error on test data or in practice.
- Loss of inductive bias.
- Increase of variance as a result of sensitivity to noise.

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Overfitting

Mitigation Strategies

Detection

- **Visualization**
  Works well for dimensionality $p \in [1, 3]$. Higher dimensionalities can only partially be visualized via projection or embedding into lower dimensionalities.

- **Validation**
  Given a validation set, the difference $Err_{val}(y) - Err_{tr}(y)$ is too large.

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Overfitting

Mitigation Strategies

Detection

- **Visualization**
  Works well for dimensionality $p \in [1, 3]$. Higher dimensionalities can only partially be visualized via projection or embedding into lower dimensionalities.

- **Validation $\rightsquigarrow$ Early stopping through model selection**
  Let m be the number of training steps, and $y_{\pi_1}, \ldots, y_{\pi_m}$ the models obtained after each step. Indicator of overfitting: The difference $\mathit{Err}_{\mathit{fit}}(y_{\pi_i}) = \mathit{Err}_{\mathit{val}}(y_{\pi_i}) - \mathit{Err}_{\mathit{tr}}(y_{\pi_i})$ increases with i.

Prevention

- **Increasing quantity or quality of the training data D**
  Quantity: More data averages out noise. Quality: Omitting poor examples enables a better fit.

- **Manually enforcing a high inductive bias**
  Choosing a hypothesis space that is less complex.

- **Regularization**
  Automatic adjustment of the loss function to penalize model complexity.

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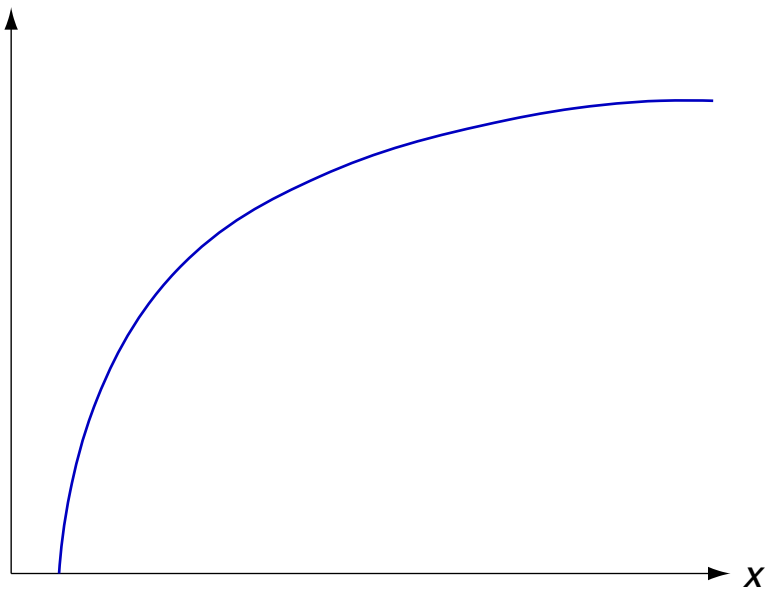
Regularization

Background: Inverse Problems



The scientific method to study physical systems:

- ➜ Forward modeling: Discovery of a model (a theory), enabling for given model parameter values to predict values of observable parameters of a system.
- ➜ Inverse modeling: Using values of observable parameters of a system to infer the values of the parameters of a model representing the system.
- ❑ Minimization of the set of model parameters that fully characterize the system.

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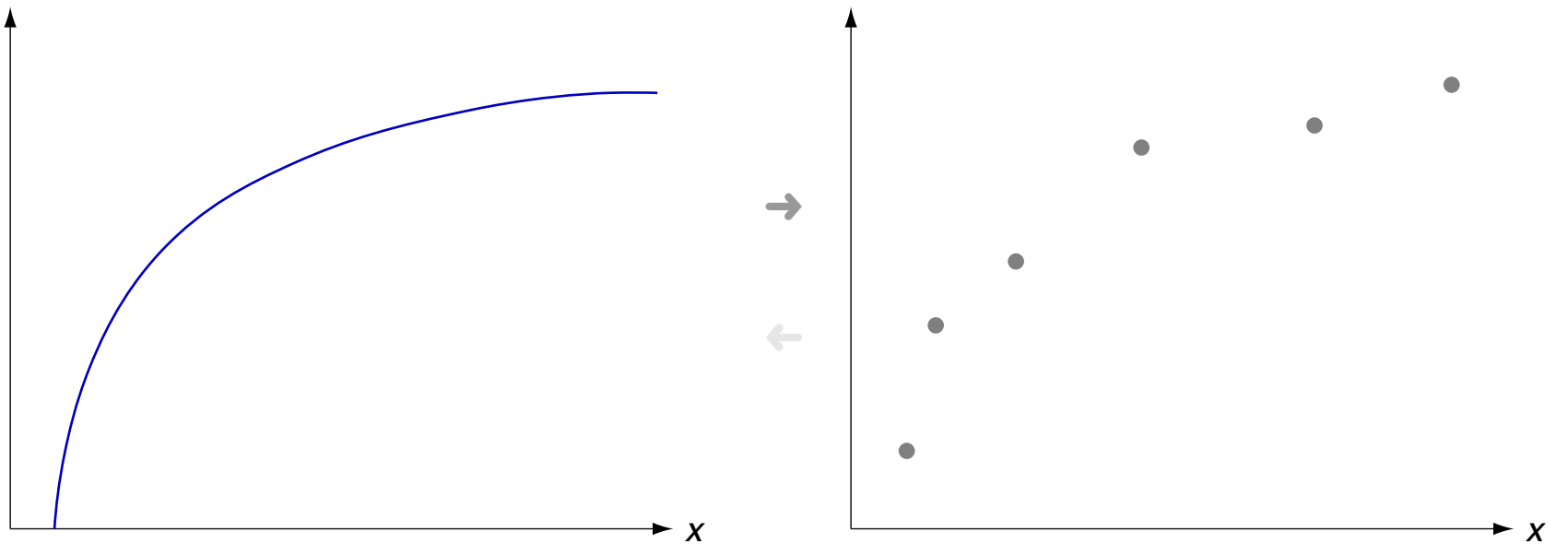

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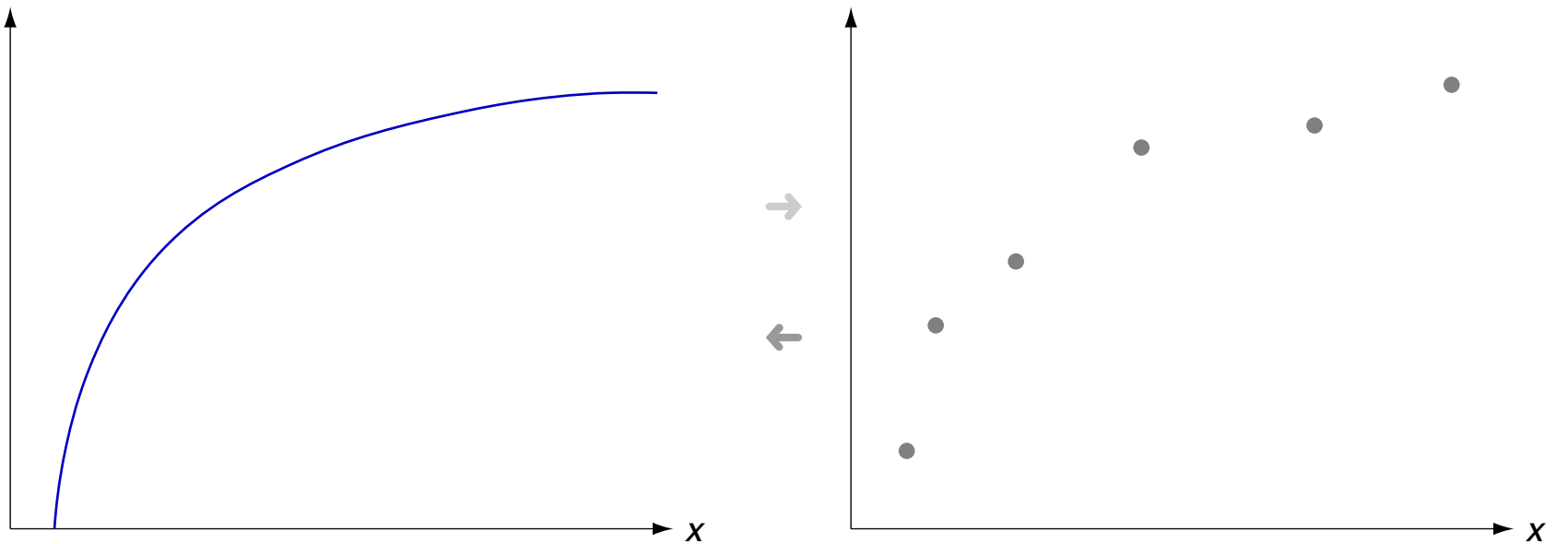

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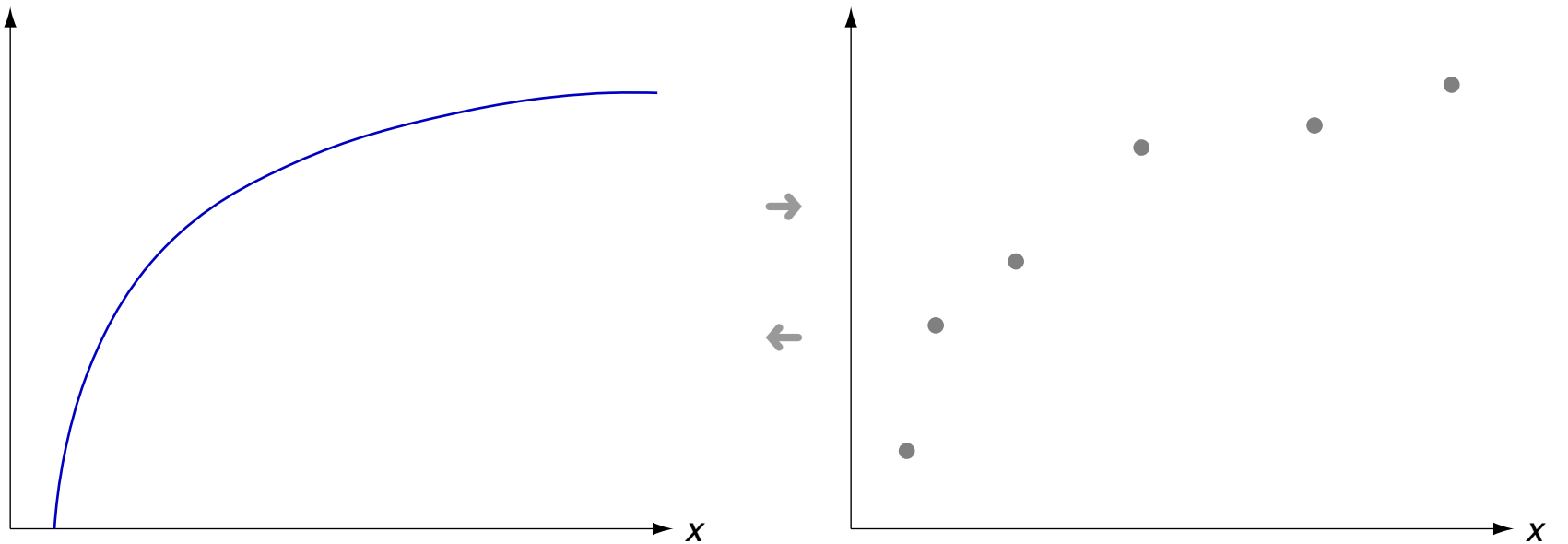

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Regularization

Background: Inverse Problems

Setting: [Wikipedia]

- The model space, a p-dimensional vector space of parameters $\mathbf{w}$ of model y.
- The data space $X \subseteq \mathbf{R}^p$ of ranges of observable system parameters.
- $Y \subseteq X$ is the range of the model functions in the model space.
- $\mathbf{y}_i$ are the predicted values of observable parameters at time / location i.
- $D = \{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_n, \mathbf{y}_n)\} \subseteq X \times Y$ are observations and model predictions.

Todo:

- Fit $\mathbf{w}$ to minimize the residuals $\mathbf{x}_i - \mathbf{y}_i$ and analyze the fit to check if model y is realistic.

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Historical inverse problems:

- Modeling the motions of the planets from observations of the night sky.
  Aristotle $\rightarrow$ Copernicus $\rightarrow$ Kepler $\rightarrow$ Newton $\rightarrow$ Einstein [NASA 2009]
- Modeling the inner composition of the Earth from observations at its surface.
- Modeling and predicting the weather based on past observations.
- ...

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Practical considerations:

- Observable parameters may be insufficient to explain the system's behavior.
- Measurements contain errors dependent on the accuracy of the instrument.
- Inverse problems are often ill-posed due to discretization.

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Regularization

Background: Ill-posed Problems

Definition 9 (Well-posed Problem [Hadamard 1902]**)**

A mathematical problem is called well-posed if

1. a solution exists,
2. the solution is unique,
3. the solution's behavior changes continuously with the initial conditions.

Otherwise, the problem is called ill-posed.

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Linear regression:

- ❑ Existence and uniqueness of a solution depends on the rank of $\mathbf{X}$.
- ➜ Use the pseudoinverse $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$.
- ❑ The solution does not depend continuously on $\mathbf{X}$.
- ❑ Introducing noise or removing data may strongly affect the solution.
- ➜ A possible way to find a solution is regularization. [Tikhonov 1977]

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Regularization

Overview

Let $L(\mathbf{w})$ denote a loss function used to optimize the parameters $\mathbf{w}$ of a model y.

Regularization introduces a trade-off between model complexity and inductive bias:

$$L_{|\mathbf{w}|}(\mathbf{w}) = L(\mathbf{w}) + \lambda R(\mathbf{w}),$$

where λ controls the importance of the regularization term $R(\mathbf{w}) \geq 0$.

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Example:

$$y(x) = w_0 + \sum_{i=1}^{6} w_i \cdot x^i; \qquad L(\mathbf{w}) = \text{RSS}(\mathbf{w})$$

$$R(\mathbf{w}) = 0w_0 + 0w_1 + 0w_2 + 1000w_3 + \ldots + 1000w_6$$

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Tikhonov regularization / ridge regression:

$$y(x) = \mathbf{w}^T\mathbf{x}; \qquad L(\mathbf{w}) = \text{RSS}(\mathbf{w})$$

$$R(\mathbf{w}) = \sum_{j=0}^{p+1} w_i^2 = \mathbf{w}^T\mathbf{w}$$

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Regularization

Regularized Linear Regression

Setting:

- The feature space $X \subseteq \mathbf{R}^p$ is an inner product space.
- $Y \subseteq \mathbf{R}$ is the output space.
- y_i is the ground truth of the feature vector $\mathbf{x}_i \in X$.
- $D = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\} \subseteq X \times Y$ is a set of examples.

Todo:

- Approximate Y, which is implicitly given via D, with a linear model function and a regularized loss function.

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Examples for linear regression problems:

- What is the credit line of a bank customer?
- What is the price of a house?

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Regularization

Regularized Linear Regression [Linear Regression]

- Given $\mathbf{x}$, predict a real-valued output under a linear model function:

$$y(\mathbf{x}) = w_0 + \sum_{j=1}^{p} w_j \cdot x_j$$

- Vector notation with $x_0 = 1$ and $\mathbf{w} = (w_0, w_1, \ldots, w_p)^T$:

$$y(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$$

- Given $\mathbf{x}_1, \ldots, \mathbf{x}_n$, assess goodness of fit as regularized residual sum of squares:

$$\text{RSS}(\mathbf{w}) = \sum_{i=1}^{n} (y_i - y(\mathbf{x}_i))^2 + \lambda \sum_{j=0}^{p+1} w_j^2 = \sum_{i=1}^{n} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda \mathbf{w}^T \mathbf{w} \qquad (1)$$

- Estimate $\mathbf{w}$ by minimizing the residual sum of squares:

$$\hat{\mathbf{w}} = \underset{\mathbf{w} \in \mathbf{R}^{p+1}}{\text{argmin}} \ \text{RSS}(\mathbf{w}) \qquad (2)$$

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Regularization

Regularized Linear Regression

- Let $\mathbf{X}$ denote the $n \times (p+1)$ matrix, where row i is $(1 \;\; \mathbf{x}_i^T)$, $\mathbf{x}_i \in D$.

  Let $\mathbf{y}$ denote the n-vector of outputs in the training set D.

$\rightsquigarrow$ $\text{RSS}(\mathbf{w}) = (\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w}) + \lambda \mathbf{w}^T\mathbf{w}$

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- Minimizing $\text{RSS}(\mathbf{w})$ by a direct method:

$$\frac{\partial\,\text{RSS}}{\partial \mathbf{w}} = -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}) + 2\lambda\mathbf{w} = 0$$

$$\begin{aligned}
\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}) - \lambda\mathbf{w} &= 0 \\
\Leftrightarrow \quad (\mathbf{X}^T\mathbf{X} + \lambda I)\mathbf{w} &= \mathbf{X}^T\mathbf{y} \\
\Leftrightarrow \quad \mathbf{w} &= (\mathbf{X}^T\mathbf{X} + \lambda I)^{-1}\mathbf{X}^T\mathbf{y}
\end{aligned}$$

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$$\begin{aligned} \mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w}) - \lambda\mathbf{w} &= 0 \\ \Leftrightarrow \quad (\mathbf{X}^T\mathbf{X} + \lambda I)\mathbf{w} &= \mathbf{X}^T\mathbf{y} \qquad \text{Normal equations.} \\ \Leftrightarrow \quad \mathbf{w} &= (\underbrace{\mathbf{X}^T\mathbf{X} + \lambda I}_{})^{-1}\mathbf{X}^T\mathbf{y} \qquad \text{If } \lambda > 0. \end{aligned}$$

Conditioning the moment matrix $\mathbf{X}^T\mathbf{X}$ [Wikipedia 1, 2, 3]

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\end{aligned}$$

Conditioning the moment matrix $\mathbf{X}^T\mathbf{X}$ [Wikipedia 1, 2, 3]

$$\hat{y}(\mathbf{x}_i) = \hat{\mathbf{w}}^T \mathbf{x}_i$$

Regression function with least squares estimator $\hat{\mathbf{w}}$.

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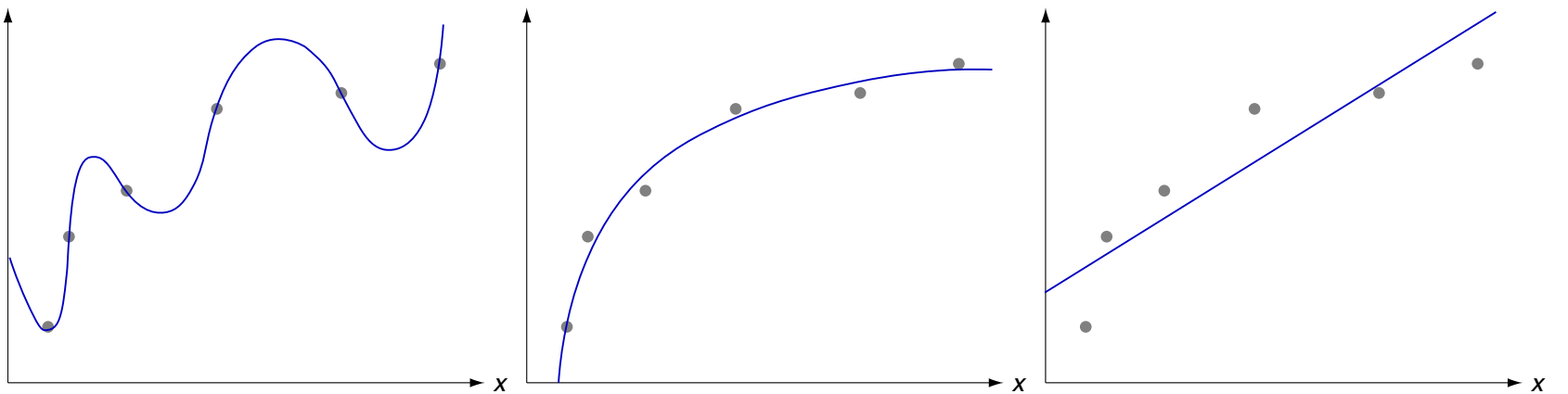
Regularization

Regularized Linear Regression: Parameter λ



$\lambda = 0 \longrightarrow$

The regularization parameter λ penalizes high weights w_i:

$$y(x) = w_0 + \sum_{i=1}^{p} w_i \cdot x^i = \mathbf{w}^T \mathbf{x}$$

$$\text{RSS}(\mathbf{w}) = \sum_{i=1}^{n} (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \lambda \mathbf{w}^T \mathbf{w}$$

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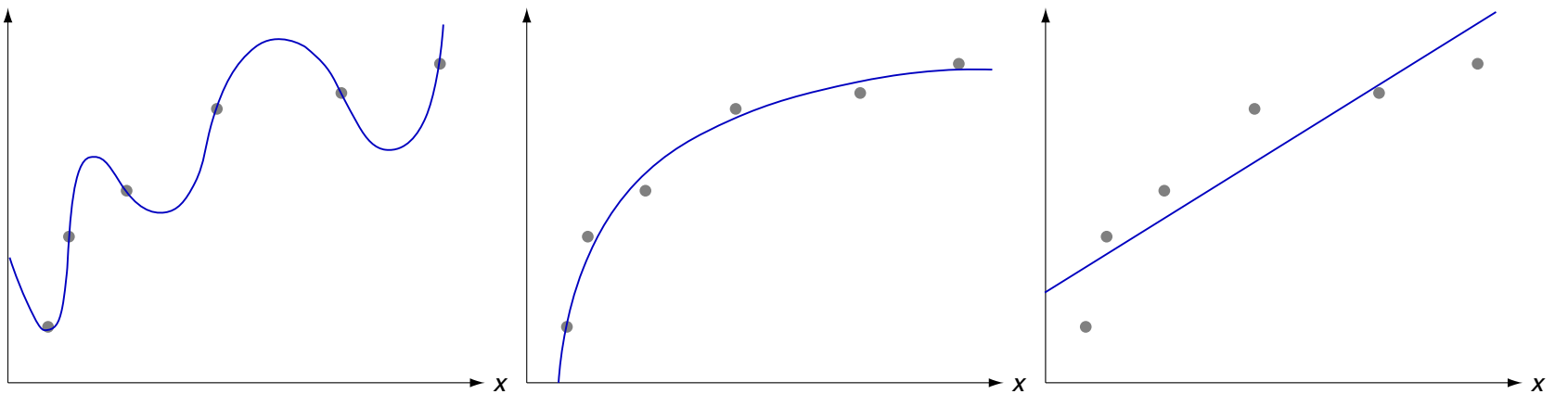
Regularized Linear Regression: Parameter λ



$\lambda = 0 \longrightarrow$

"No black-box procedures for choosing the regularization parameter λ are available, and most likely will never exist." [Hansen and Hanke 1993]

Heuristics:

- Model selection for a choice of $\lambda_1, \ldots, \lambda_m$. [StackOverflow 2017]
- More advanced heuristics: [Bauer and Lukas 2011]

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Remarks:

- What is underfitting?
- The parameter w_0 is sometimes not included in the regularization term so that non-normalized data can still be fitted for low values of λ.
- In statistics, the method is known as ridge regression, in machine learning it is known as weight decay, and with multiple independent discoveries, it is also variously known as the Tikhonov-Miller method, the Phillips-Twomey method, the constrained linear inversion method, and the method of linear regularization. It is related to the Levenberg-Marquardt algorithm for non-linear least-squares problems. [Wikipedia]
- The regularization term renders the factor $(\mathbf{X}^T\mathbf{X} + \lambda I)$ invertible, even if the number of features p exceeds the number of examples n in D.
- Hadamard thought that ill-posed problems are a pure mathematical phenomenon and that all real-life problems are "well-posed." However, in the second half of the century a number of very important real-life problems were found to be ill-posed. In particular, ill-posed problems arise when one tries to reverse the cause-effect relations: to find unknown causes from known consequences. Even if the cause-effect relationship forms a one-to-one mapping, the problem of inverting it can be ill-posed. [Vapnik 2000]
- According to Hadamard's philosophy, ill-posed problems are actually ill-posed, in the sense that the underlying model is wrong.
- Supervised learning has been shown to be an inverse problem. [de Vito 2005]

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